

Eric C. Blow

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Education

Princeton University, Princeton, N.J.

Doctor of Philosophy, Ph. D, Electrical Engineering, Integrated Photonics, 2023

Thesis Title: Microwave Photonic Cancellation: RF Analysis, III-V and Silicon Integration, Development of Balanced and Hybrid Architectures

Topics: Microwave Photonics, Neuromorphic Photonics, InP Integration, Si Integration, SiN Integration, Packaging

Relevant Online Material: [Thesis](#) | [Code](#)

Master's Degree, Electrical Engineering, Photonics and Electronic Devices, 2017

Thesis Topic: Microwave Photonic Cancellers

Relevant Coursework: Photonic Systems, Silicon Photonics, Nanophotonics, Microelectronics, and Nano Fabrication

Teaching Assistant: Photonic Systems, Silicon Photonics, Lightwave Communication, and Intro to Engineering

Guest Lecturer (2017-2026): Silicon Photonics, Photonics and Light Wave Communication

The College Of New Jersey, Ewing, N.J.

Bachelor of Science, B.S., Electrical and Computer Engineering, 2015

Concentration: Wireless communications, RF electronics, and Nonlinear photonics.

Awards: Merit Scholarship, Armstrong Scholar Award - GPA Award, Conrad Johnson Award – Senior Thesis Award, Mayo Business Plan Winner

Engineering Research and Industry Experience

NEC Laboratories America: Princeton, NJ

Researcher within Optical Networking and Sensing Group: 07/2023-Present

- Research lead in the development of Integrated Photonic Systems for AI Accelerators, RF Analog Signal Processing, Integrated Photonic Neural Networks, Fiber-based Photonic AI Computing
- Architected integrated photonic systems spanning Silicon, SiN, and InP platforms through DARPA LUMOS tapeouts.
- Contributed to research programs within the Fiber Sensing and Photoacoustic Sensing by integrated existing systems.
- Published 10+ papers and delivered invited presentations at major conferences including IPC, SPIE, and COOL Chips.
- Presented research internally and externally at NEC Open House '24 and '25, in Tokyo, Japan to C-Suite Executives, key customers, and government representatives
- Research presented by CTO Motoo Nishihara-san at NEC Innovation Day '25

Princeton University: Lightwave Laboratory, Princeton, NJ

Undergraduate Researcher: 2014-2015

Graduate Researcher: 2015-2023

Visiting Research Collaborator: 2023-Present

Research Advisor: Dr. Paul R. Prucnal

Focus: Microwave Photonics, Neuromorphic Photonics, Cancellation Systems

- Developed Novel Architecture for RF Optimization: Balanced Photonic Canceller and Hybrid Photonic Canceller
- Integrated Cancellers on Indium Phosphide, active and passive Silicon, and Silicon Nitride

- Designed, simulated, fabricated, and characterization of Integrated microwave photonic systems
- Lead commercial integrated photonic foundry MPW tapeouts: Indium Phosphide (JePPiX), Passive Silicon (ANT), Silicon Nitride (LioniX), and Active Silicon (AMF)
- Trained in Lithography and numerous cleanroom microelectronic/photonic fabrication processes
- Gained valuable experiences in software development, machining, lab automation
- Designed and Simulated Micro Ring Resonators, Electroabsorption Modulators, Semiconductor Optical Amplifiers, Mach-Zender Modulators, Michelson-Morley Modulators, DFB Lasers, Balanced Photodetectors, Multimode Interferers and other integrated RF photonic devices
- Wrote 6+ successful government and private research grants which resulted over a million USD in funding
- Authored 2 book chapters and 20+ academic papers within the field of photonics

Bascom Hunter Technologies, Baton Rouge, LA

Defense Contractor

Senior Photonics Engineer: Full Time 2019 – 2021 / Part Time: 2021 - 2023

Focus: Neuromorphic Photonics, High Dynamic Range Photonic Links, and Microwave Photonic Cancellers

- Authored 7 successful government research proposals resulting in 10 million USD of R&D funding
 - Awarded multiple SBIR Phase I and Phase II programs and one Phase III program
- **Principle Investigator** for Navy program N203-149: High Dynamic Range Microwave Photonic Link for LightRIDR System
- **Principle Investigator** for Navair program N202-099: NOHP-Neuromorphic Optoelectronic Hybrid Processor
- **Lead Engineer** for Army program A20B-T004-0271: Massively Parallel Photonic Artificial Neural Network Accelerator
- **Photonic Engineer** for SOCOM program S211-003: Wideband and Analog Radio Frequency Fingerprinting At a Distance
- **Photonic Engineer** for Navy Phase III SBIR N112-170: Future Advanced Satellite Communication (SATCOM) Technologies (FAST) SATCOM Pre-Planned Product Improvement (P3I) for Navy Multiband Terminal (NMT) (SPIN)
- **Photonic Engineer** for NASA program Z2.02-5311: HPSC-compatible Neuromorphic Photonic Coprocessor
- **Photonic Engineer** for Strategic Capabilities Office SCO 213-003: Photonic AI Processor (PHASOR)
- Designed and constructed integrated photonic testing facilities

The College of New Jersey: Photonics Laboratory, Ewing, NJ

Nonlinear Photonics Research Laboratory

Student Researcher 2012-2015

Research Advisor: Dr. David J. McGee

- Research focused on linear and nonlinear optical techniques to probe and characterize novel optoelectrical materials
- Studied optical and electrical properties of Chromophore-Functionalized Carbon Nanotubes
- Responsible for conducting numerous experiments including optical switching, birefringence, and surface relief grating
- Conducted second harmonic generation experiments to determine orientational order of organic dimolecule

Naval Surface Warfare Center Carderock Division, Philadelphia, PA

Science Engineering Apprenticeship Program (SEAP): Engineering Intern , 2011

Summer Temporary Employed Position (STEP): Electrical Engineering Technician (05/2012-08/12; 05/2013-08/2013)

Skills

Technical:

- | | |
|--|---|
| • Circuit Design Software (PSPICE) | • Automation Software (Python, LabVIEW) |
| • 3D Modeling Software (Solidworks) | • Mathematical Modeling (MATLAB, Mathematica) |
| • Software Languages (C++, VB, Python) | • Microwave/Communication Design (AWR Microwave Office) |
| • Photonic Simulation (PICWave) | • Photonic Integrated Circuit Design (Optodesigner) |
| • PCB Design (Altium, Eagle) | • Photonic Layout Tools (ZeroPDK, GDS Factory) |

Conferences

2016 - IEEE Microwave Photonic Conference
2018 - IEEE Photonics Conference
2020 - Conference on Laser and Electro-Optics (CLEO)
2021 - SPIE Photonic West [invited]
2022 - International Conference on Physical Computing [invited]
2023 - IEEE Photonic Conference [invited]
2024 - SPIE Photonic Europe
2025 - IEEE Photonic Conference [committee member, invited, contributed x2]
2026 - IEEE COOL Chips [Keynote]
2026 - Wendy and Eric Schmidt Meeting on Unconventional Computing [invited]

Publications

- Reviewer for IEEE, Journal of Lightwave Technology, Photonic Technology Letters, Nature, Optica

- 1) A. Tait, M. Nahmias, B. Shastri, M. Chang, A. Wu, E. Zhou, **E. C. Blow**, T. Ferreira de Lima, B. Wu, and P. R. Prucnal. "Balanced WDM weight banks for analog optical processing and networking in silicon." In *Summer Topicals, IEEE/OSA*, number MC2.3 (Invited), July 2015.
- 2) M. P. Chang, J. Sun, **E. C. Blow**, and P. R. Prucnal, "A Microwave Photonic Canceller for Improved Interference Rejection in Full Duplex Radio", IEEE Photonic Conference, 2016
- 3) M. P. Chang, J. Sun, **E. C. Blow**, M. Lu, and P. R. Prucnal, "Sensors for Everyday Life, Chapter 10: Optical-Based Interference Cancellation in Wireless Sensor Networks" Volume 23 of the series Smart Sensors, Measurement and Instrumentation pp 273-301, Nov 2016
- 4) **E. C. Blow**, M. P. Chang, P. R. Prucnal, "Microwave Photonic Interference Canceller: Noise Figure Reduction via Balanced Architecture", MPC, Oct. 2016
- 5) P. Y. Ma, Y. Huang, M. P. Chang, **E. C. Blow**, S. Zhang, P. R. Prucnal, "Microwave Photonic Chaos Based Device Fingerprinting", OFC, 2016
- 6) M. P. Chang, **E. C. Blow**, J. Sun, M. Lu, P. R. Prucnal, "Integrated Microwave Photonic Circuit for Self-interference Cancellation", IEEE Transaction Theory and Techniques, 2017
- 7) M. P. Chang, **E. C. Blow**, J. Sun, M. Lu, P. R. Prucnal, "RF Characterization of an Integrated Microwave Photonic Circuit for Self-Interference Cancellation", IEEE Transaction Theory and Techniques, 2017
- 8) **E. C. Blow**, M. P. Chang, and P. R. Prucnal, "**Microwave Photonic Canceller: Noise Reduction Via Balanced Architecture**", IEEE Transactions on Microwave Theory and Techniques 2022
- 9) **Blow, Eric C.**, Prannay Kaul, and Paul R. Prucnal. "Integrated Balanced Microwave Photonic Canceller." 2018 IEEE Photonics Conference (IPC). IEEE, 2018.
- 10) Tait, Alexander N., **E. C. Blow**, et al. "Demonstration of multivariate photonics: blind dimensionality reduction with integrated photonics." *Journal of Lightwave Technology* (2019).
- 11) Huang, Chaoran, **E. C. Blow**, et al. "Accelerated secure key distribution based on localized and asymmetric fiber interferometers." *Optics express* 27.22 (2019): 32096-32110.
- 12) Huang, Chaoran, **E. C. Blow**, et al., "Demonstration of Multi-Channel Feedback Control for On-Chip Microring Weight Banks" OFC Post-deadline Paper
- 13) Huang, Chaoran, **E. C. Blow**, et al., "Demonstration of scalable microring weight bank control for large-scale photonic integrated circuits." *APL Photonics* 5.4 (2020): 040803. **E. C. Blow**, et al. "Silicon Photonic Weights for Microwave Photonic Canceller", CLEO 2020
- 14) T. F. de Lima, **E. C. Blow**, B. J. Shastri, and P. Prucnal, "Real-time Operation of Silicon Photonic Neurons," in Optical Fiber Communication Conference (OFC) 2020, OSA Technical Digest (Optical Society of America, 2020), paper M2K.4.
- 15) C. Huang, **E. C. Blow**, et al "Demonstration of photonic neural network for fiber nonlinearity compensation in long-haul transmission systems," in Optical Fiber Communication Conference. Optical Society of America, 2020, pp. Th4C–6 (Postdeadline)

- 16) **E. C. Blow**, et al, "Silicon Photonic Weights for Microwave Photonic Cancellor", CLEO 2020
- 17) **E. C. Blow**, Paul R. Prucnal, "In-Band Full-Duplex Technologies and Applications, Chapter: IBFD Optical Systems" Artech House 2021
- 18) PATENT "Light-emitting diode neuron", S Bilodeau, P Prucnal, M Hack, TF de Lima, HT Peng, E Blow, US Patent App. 17/125,287
- 19) Huang, Chaoran, Shinsuke Fujisawa, Thomas Ferreira de Lima, Alexander N. Tait, **Eric C. Blow**, Yue Tian, Simon Bilodeau et al. "A silicon photonic–electronic neural network for fibre nonlinearity compensation." *Nature Electronics* 4, no. 11 (2021): 837-844.
- 20) Zhang, Weipeng, Chaoran Huang, Simon Bilodeau, Aashu Jha, Eric Blow, Thomas Ferreira De Lima, Bhavin J. Shastri, and Paul Prucnal. "Microring Weight Banks Control beyond 8.5-bits Accuracy." *arXiv preprint arXiv:2104.01164* (2021).
- 21) De Lima, Thomas Ferreira, Chaoran Huang, Simon Bilodeau, Alexander N. Tait, Hsuan-Tung Peng, Philip Y. Ma, **Eric C. Blow**, Bhavin J. Shastri, and Paul Prucnal. "Real-time Operation of Silicon Photonic Neurons." In *Optical Fiber Communication Conference*, pp. M2K-4. Optical Society of America, 2020.
- 22) **E. C. Blow**, T. Ferreira de Lima, H-T Peng, W. Zhang, C. Huang, B. J. Shastri, and Paul R. Prucnal "Broadband Radio-Frequency Signal Processing with Neuromorphic Photonics", SPIE Photonic West (invited) 2022.
- 23) Hsuan-Tung Peng, Thomas Ferreira de Lima, **E. C. Blow**, S Bilodeau, Aashu Jha, Chaoran Huang, Bhavin Shastri, and P. R. Prucnal "Time Series Prediction and Classification using Silicon Photonic Neuron with a Self-Connection", CLEO 2022
- 24) Thomas Ferreira de Lima, **Eric C. Blow**, Eli A. Doris, Simon Bilodeau, Weipeng Zhang, Aashu Jha, Hsuan-Tung Peng, Chaoran Huang, Alexander N. Tait, Bhavin J. Shastri, and Paul R. Prucnal "Design Automation of Photonic Resonator Weights", De Gruyter Journal 2022
- 25) Peng, Hsuan-Tung, et al. "Time Series Prediction and Classification using Silicon Photonic Neuron with a Self-Connection." CLEO: Science and Innovations. Optica Publishing Group, 2022.
- 26) Zhang, W., Tait, A., Huang, C., Ferreira de Lima, T., Bilodeau, S., Blow, E.C., Jha, A., Shastri, B.J. and Prucnal, P., 2023. Broadband physical layer cognitive radio with an integrated photonic processor for blind source separation. *Nature Communications*, 14(1), p.1107
- 27) **Eric C. Blow**, Yusuf Jha, Lei Xu, Weipeng Zhang, Gary Xu, and Paul R. Prucnal "Hybrid Microwave Photonic Cancellor for self-interference cancellation", IEEE Microwave Theory and Technique 2023
- 28) **Eric C Blow**, Simon Bilodeau, Weipeng Zhang, Thomas Ferreira de Lima, Josh Lederman, Paul R Prucnal "Radio-Frequency Linear Analysis And Optimization Of Silicon Photonic Neural Networks", ARP 2024 (Invited)
- 29) Joshua C. Lederman, Weipeng Zhang, Thomas Ferreira de Lima, **Eric C. Blow**, Simon Bilodeau, Bhavin J. Shastri, Paul R. Prucnal "Real-Time Blind Photonic Interference Cancellation for mmWave MIMO", *Nature Communication*, 2023
- 30) Joshua C. Lederman, Yusuf Jimoh, Weipeng Zhang, Thomas Ferreira de Lima, **Eric C. Blow**, Simon Bilodeau, Bhavin J. Shastri, Paul R. Prucnal , A Multi-Layer Topologically Reconfigurable Broadcast-and-Weight Photonic Neural Network, IPC 2023
- 31) T. F. de Lima, J. C. Lederman, W. Zhang, J. Hu, E. C. Blow, K. Asahi, Y. Aono, T. Wang, and P. R. Prucnal, "Real-Time Blind Source Separation with Integrated Photonics for Wireless Signals," in CLEO 2023
- 32) E.C. Blow et. Al. "Link Loss Analysis of Integrated Linear Weight Bank within Silicon Photonic Neural Network, SPIE Europe 2024
- 33) Lederman, Joshua C., E.C. Blow, et al. "Low-latency passive thermal desensitization of a silicon micro-ring resonator with self-heating." *APL Photonics* 9.7 (2024).
- 34) Eric C. Blow, et al., "Neuromorphic Photonics-Enabled Near-Field RF Sensing with Residual Signal Recovery and Classification", IPC 2025
- 35) Giovanni Milione, Eric C. Blow, et al., "Machine Learning, Distributed Acoustic Sensing, and Critical Infrastructure Monitoring", IPC 2025 (Invited)
- 36) Yuxin Wang, Eric C Blow, et al. "Scalable Photonic Neurons for High-Speed Automatic Modulation Classification", IPC 2025
- 37) Lederman, Joshua C., Eric C Blow, et al. "Integrated optical-to-optical gain in a silicon photonic modulator neuron." *Optica* 12.12 (2025): 1904-1911.
- 38) Doris, Eli A., Eric C. Blow, Hyuma Umeda, Jiawei Zhang, and Paul R. Prucnal. "Emerging integrated photonic technologies leveraging multimaterial integration for AI and datacenter applications." *Optics Express* 33, no. 20 (2025): 41633-41652.

References are available upon request